

ENCODER (FROZEN)

DECODER

PREDICTION HEADS

OUTPUT

SensorMultiHeadDecoder Architecture

Per-Row G-Code Recovery on a Frozen Multi-Modal Sensor Encoder · ~39.9M total parameters

64-second Sensor Window · T=256 @ 4 Hz · C=98 channels · 7 modality groups

accel (18)

gyro (18)

magneto (18)

temp (6)

color (24)

audio-RMS (6)

motor-current (8)

from 6 distributed sensor units (frame_l2, frame_l3, frame_r2, spindle2, y_bed__3, y_bed__4) · shape (B, 256, 98)

MM-DTAE-LSTM Encoder (8.6M params, FROZEN)

prior contribution; not retrained in this paper

Per-Modality MLPs

7 projections → d=256

LN + GELU + Dropout

+ learned modality gates

Cross-Modal MHA

multi-head self-attention

denoising objective

σ-Gaussian input noise

2-Layer LSTM

hidden = 256

temporal accumulation

outputs memory tensor

FROZEN

96.3% 9-class

operation accuracy

(prior work)

Encoder Memory $M \in 256 \times 256$

consumed by decoder cross-attention; not re-encoded per AR step

Transformer Decoder (31.28M params)

8 layers, d_model = 384, 12 heads/layer, GELU FFN, dropout 0.1, pre-LayerNorm

Decoder Layer ℓ (× 8 stacked)

Masked Self-Attention

causal mask over $y_{\{1..t-1\}}$

Cross-Attention

Q = hidden, K, V = memory M

Feed-Forward (GELU)

d_ff = 1536, dropout 0.1

residual + LayerNorm after each sublayer (pre-norm)

Token Embedding + Positional Encoding

vocab = 2,418 · learned sinusoidal positions

Scheduled Sampling (training)

p_max swept $\in \{0, 0.25, 0.5, 0.75\}$; SS=0 deployed

Decoder Hidden State $h_t \in 384$

shared across all six heads

Token Head

PRIMARY · inference

2,418-class softmax

over full vocabulary

bigram + FSM mask applied

Type Head

auxiliary, multi-task

4-way:

SPECIAL / CMD /

PARAM / NUMERIC

Motion-Command

auxiliary, multi-task

6-way:

G0 / G1 / G2 / G3

G53 / M30

Parameter-Type

auxiliary, multi-task

10-way:

X / Y / Z / I / J / K

R / F / S / P

Sign Head

auxiliary, multi-task

3-way:

POS / NEG /

ABSENT

Per-Position Digit

auxiliary, regression-style

6 positions ×

11 classes

(digits 0–9 + PAD)

token logits

Grammar-Constrained Decoding (inference time only)

Bigram Mask M_b

$|V|=2,418 \rightarrow$ avg 31.8 valid next tokens

FSM Mask

G0/G1 forbid R/I/J; M30 forbids continuation

Apply to token logits

$\ell_t \leftarrow \ell_t \cdot M_b[y_{\{t-1\}}, :] \cdot \text{FSM_mask}(s_t)$

BOS G1 X NUM_X_1492 Y NUM_Y_1485 F NUM_F_22 EOS